Journal of Intelligent Manufacturing (2025) 36:4785–4806
https://doi.org/10.1007/s10845-024-02475-3

A multi-task effectiveness metric and an adaptive co-training method for enhancing learning performance with few samples

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Received: 6 October 2023 / Accepted: 26 July 2024 / Published online: 12 August 2024
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Abstract
The integration of deep learning (DL) into vision inspection methods is increasingly recognized as a valuable approach to substantially enhance the adaptability and robustness. However, it is well known that high-performance neural networks typically require large training datasets with high-quality manual annotations, which are difficult to obtain in many manufacturing processes. To enhance the performance of DL methods for vision task with few samples, this paper proposes a novel metric called Effectiveness of Auxiliary Task (EAT) and presents a multi-task learning approach utilizing this metric for selecting effective auxiliary task branch and adaptive co-training them with main tasks. Experiments conducted on two vision tasks with few samples show that the proposed approach effectively eliminates ineffective task branches and enhances the contribution of the selected tasks to the main task: reducing the average normalized pixel error from 0.0613 to 0.0143 in pose key-points detection and elevating the Intersection over Union (IoU) from 0.6383 to 0.6921 in surface defect segmentation. Remarkably, these enhancements are achieved without necessitating additional manual labeling efforts.

Keywords Machine vision · Deep learning · Multi-task learning · Few samples · Auxiliary task

Introduction

With the development of information technology and advanced manufacturing, machine vision technology is increasingly integrated into modern manufacturing processes. The implementation of vision tasks such as surface defect detection (Fan et al., 2023) and monocular pose estimation (Usamentiaga et al., 2022) provides a foundation for the automation of complex procedures like quality control and product assembly in manufacturing processes.

In recent years, the integration of deep learning (DL) into vision inspection methods is gradually becoming a key to significantly improve the adaptability and robustness of traditional vision inspection. However, it is well known that high-performing neural networks typically require large training datasets with high-quality manual annotations. The intricacy of product features, the necessity for industrial domain expertise, and the requirement for precise labeling make the annotation process for many vision tasks labor-intensive and time-consuming (as depicted in Fig. 1), resulting in only few samples being collected for them (Amhaz et al., 2016; Song et al., 2014; Tabernik et al., 2020). DL methods struggle to achieve satisfactory performance on such vision tasks (Wei et al., 2020), which poses a significant challenge for the wider adoption of deep learning in manufacturing vision applications.

In the realm of machine vision, various strategies including data augmentation, domain adaptation, and multi-task learning are used for the challenges posed by few samples. Data augmentation automates the generation of new labeled samples through generator like GAN generation (Zhao et al., 2018) and NeRf rendering (Ge et al., 2022). Based on skillfully designed generators and high computational cost, data augmentation can expand labeled samples to improve the performance of deep learning models. Domain adaptation transfers knowledge from domains with sufficient supervisory signals to task domains (like from product digital model

This work was supported by the National Natural Science Foundation of China (No. 52375478) and National Defense Basic Scientific Research Program of China (No. JCKY2021602B021).

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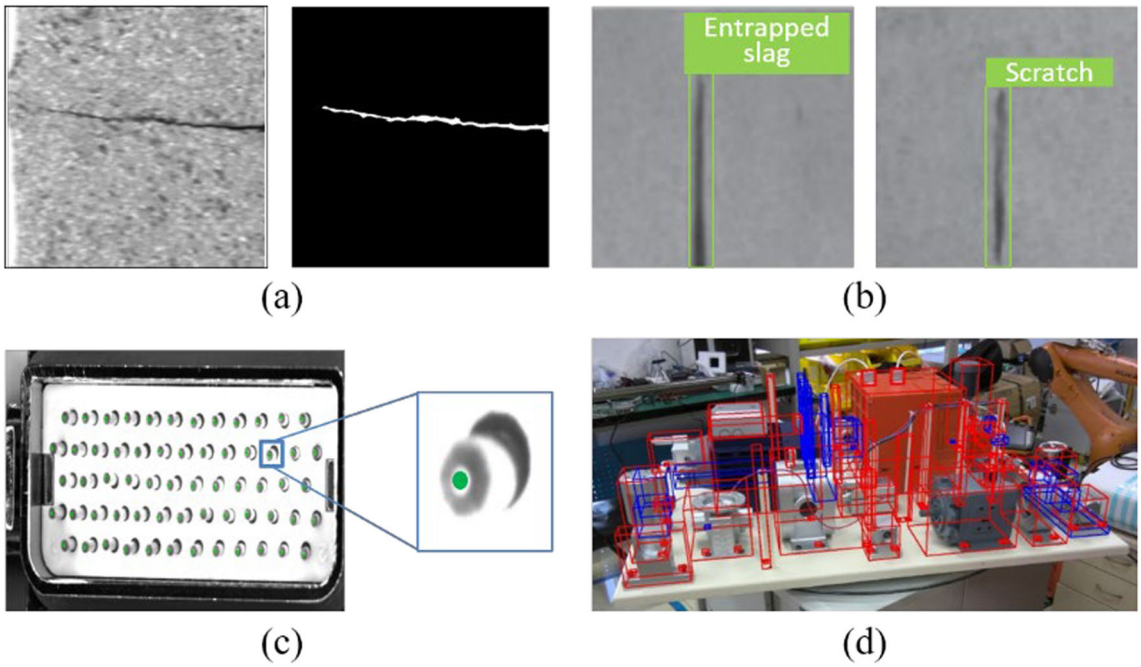

Fig. 1 Examples of vision tasks with few samples. **a** Commutator surface defect segmentation (Tabernik et al., 2020), which needs to label the precise location of cracks pixel by pixel. **b** Surface defect detection of steel (He et al., 2020), which can be correctly distinguished and labeled by only few professionals due to the high similarity of surface defects such as *entrapped lag* and *scratch*. **c** Electrical connector pin position detection (Zhao et al., 2022), which requires empirical estimation of the vertices of many pins and pixel-level precise annotation of them. **d** The 6D pose estimation of the product during the assembly process, which requires precise axis-aligned bounding box labeling of many products

domain to product physical domain). Successful applications of domain adaptation on tasks like key-points detection (Jiang et al., 2021) and target detection (Wang et al., 2022a) demonstrate that neural networks can effectively reduce their dependence on large annotated dataset through skillfully designed knowledge transfer processes.

Multi-task learning is a machine learning paradigm that simultaneously trains multiple tasks to improve the performance of a single task by effectively sharing information between tasks. Unlike data augmentation and domain adaptation, multi-task learning does not depend on skillfully designed generators or knowledge transfer processes, making it a versatile approach that can be flexibly applied to various tasks. The crux of the multi-task learning lies in the selection of suitable task branches for training. Appropriate auxiliary task branches can enhance the performance of the main task, while inappropriate auxiliary task branches may lead to a decline in the main task's performance.

Researchers (Guo et al., 2020; Lee et al., 2019; Li et al., 2021) typically rely on their experiential knowledge to select task branches that are related to the main task as auxiliary task branches. However, this approach heavily depends on individual expertise, and the effectiveness of chosen auxiliary task branches in improving the performance of the main task remains uncertain. To address this challenge, various automatic selection methods for auxiliary task branches have been proposed (Chen et al., 2018; Du et al., 2018; Fifty et al., 2021; Liu et al., 2022). Nevertheless, these methods solely evaluate the impact of auxiliary task branches on the main task based on training error. For vision tasks with few samples, overfitting hampers the accurate reflection of generalization performance changes in DL methods solely through training error. Consequently, the efficacy of the auxiliary task branches selected by these methods cannot be guaranteed when only few samples are available.

Furthermore, the influence of auxiliary task branches on main task may be different at different stages of training process. The same auxiliary task branch may improve the performance of the main task during certain stages, while potentially hindering performance in other stages (Du et al., 2018). Effectively incorporating the information from auxiliary task branches at the appropriate timing is a significant challenge in the successful implementation of multi-task learning approaches.

Motivated by aforementioned challenges, a novel evaluation metric termed as the Effectiveness of Auxiliary Task (EAT) is proposed in this paper. EAT systematically evaluates the influence of auxiliary task branches on the performance of the main task, utilizing semantic correlations observed within

the training set and the impact of auxiliary task information on the validation set. Building upon this metric, two essential method are formulated: an automatic selection method and an adaptive co-training method. Leveraging these foundations, a novel multi-task learning approach is proposed to enhance the learning performance for vision tasks with few samples.

The contribution of this paper can be summarized as follows:

(1) A novel multi-task learning approach is proposed, which takes vision task as the main task and automatically select auxiliary task branches to perform adaptive co-training. It is applicable to various neural networks and has the potential to improve their performance for vision tasks with few samples, without requiring additional manual annotation.
(2) An evaluation metric, termed the Effectiveness of Auxiliary Task (EAT), is proposed to evaluate the impact of auxiliary task branches on the main task.
(3) An automatic select method and an adaptive co-training method are proposed for auxiliary branches' automatic selection problem and information sharing timing problem.

The rest of this paper is organized as follows. Related works about this paper are given in section "Related works". Next, proposed method is narrated thoroughly in section "Methodology and design". Afterwards, section "Experiments" describes verification of proposed method for two vision tasks with few samples and corresponding discussions. Finally, section "Conclusion" concludes this paper.

Related works

Few-shot learning and multi-task learning

Machine learning problems with few samples are often categorized as few-shot learning, and the few-shot learning is defined as a type of machine learning problems (specified by experience E, task T and performance measure P), where E contains only a limited number of examples with supervised information for the target T (Wang et al., 2020). Various techniques such as data augmentation (Ge et al., 2022; Wang et al., 2022b), transfer learning (Jiang et al., 2021; Kang et al., 2023), meta-learning (Cheng et al., 2021; Lee et al., 2023), and multi-task learning (Bouniot et al., 2022) are considered to have the potential to address few-shot learning (Song et al., 2023; Zeng & Xiao, 2024). Data augmentation often require skillfully designed generators and high computational cost to solve the few-samples problem. Whereas, the success of both transfer learning and meta-learning needs to be supported by a suitable knowledge transfer process. In contrast to these, multi-task learning does not depend on skillfully designed generators or knowledge transfer processes, making it a versatile approach that can be flexibly applied to various tasks.

Multi-task learning (MTL) is a machine learning paradigm which aims to improve model generalization by leveraging domain-specific information contained in the training signals of related tasks (Vandenhende et al., 2022). MTL can be categorized into two types: hard parameter sharing and soft parameter sharing. In hard parameter sharing, different task branches task branches share common underlying parameters while also possessing their own task-specific parameters (Chen et al., 2018; Kendall et al., 2018; Kokkinos & IEEE, 2017). An exemplary instance of hard parameter sharing is UberNet (Kokkinos & IEEE, 2017), where vision tasks at multiple levels are collaboratively processed through shared parameters. Conversely, soft parameter sharing enables different task branches to establish mutual constraints between their parameter spaces via regularization methods without directly sharing underlying parameter (Gao et al., 2019; Liu et al., 2019; Ruder et al., 2019). Cross-stitch network (Misra et al., 2016) is the first attempt to introduce soft parameter sharing into deep learning.

Generally, task-specific parameters are increased by soft parameter sharing, making it well-suited for scenarios involving substantial differences between tasks. However, compared to hard parameter sharing, soft parameter sharing carries a higher risk of overfitting due to the increased parameters. On the other hand, hard parameter sharing is better suited for situations where task variability is limited, as its smaller parameter size helps to mitigate the risk of overfitting. This makes hard parameter sharing more suitable for tasks with only few samples.

Design and selection of task branches

Corresponding labeled samples are required by different task branch of multi-task learning model. Increased data costs are incurred with a greater number of branches. One approach to address this issue is the design of auxiliary task branches, such as image reconstruction, as self-supervised learning tasks (Gui et al., 2022; Lu et al., 2020). Alternatively, annotation data can be generated automatically based on domain knowledge and existing annotated data. Subsequently, auxiliary task branches are formulated based on the generated annotation data. Foreground–background segmentation annotation and closeness annotation can be generated by existing target detection annotation (Lee et al., 2019), and picture action classification annotation can be generated by existing video action detection annotation (Li et al., 2021). In comparison to the method of designing self-supervised learning task branches, the method of generating annotations

through domain knowledge exhibits increased flexibility and applicability across a broader spectrum of tasks.

The selection of appropriate task branches for co-training is a crucial foundation of multi-task learning. Previous studies (Kokkinos & IEEE, 2017; Li et al., 2021) have predominantly relied on empirical methods to identify relevant task branches. In order to mitigate the dependence on researchers' expertise and select appropriate task branches for co-training, a method proposed by Fifty C (Fifty et al., 2021) determines whether two tasks should be co-trained based on the reduction of training error observed after co-training. Other researchers have proposed approaches (Du et al., 2018; Liu et al., 2022) that assess the influence between task branches by analyzing the change in parameter gradient resulting from the training loss of different task branches. Furthermore, Du et al. (2018) has highlighted that the influence between two task branches is not fixed. At certain stages of the training process, these task branches may exhibit a mutually reinforcing effect, while at other stages, they may exert a detrimental impact on each other.

Although the aforementioned studies and their subsequent works (Chen et al., 2022; Yan et al., 2022) have made valuable contributions to the field of task branch selection, they have primarily assessed the influence between different task branches based on the performance of tasks on the training set. When an ample number of samples are available, the data distribution of the training set generally aligns with that of the real task, and enhancements in model performance on the training set can effectively indicate improvements in model generalization. However, when confronted with few samples, the training set becomes inadequate in characterizing the data distribution of the real task. Consequently, relying solely on training set information to evaluate the influence between task branches is no longer a reliable approach. Thus, the evaluation of the impact of auxiliary task branches on the main task in scenarios with few samples remains a challenging problem.

Methodology and design

Approach overview

In the proposed multi-task learning approach, vision tasks with few samples are regarded as main task and are combined with auxiliary task branches to formulate a multi-task learning model of hard parameter sharing. There are four steps in the proposed approach. A visual depiction of them is shown in Fig. 2. The details of the auxiliary task branches and their labels are discussed in section "Experiments".

Effectiveness of auxiliary task

Effectiveness of auxiliary task (EAT) is the proposed metric to measure whether auxiliary task branches have improved main task performance in the current training stage. Inspired by the network attention-region analysis method in Grad-CAM (Selvaraju et al., 2017), the similarity of shared feature maps gradients resulting from different task training losses is computed to characterize the degree of semantic correlation between task branches. The success of Grad-CAM inspires us that if the gradient distributions of the losses of two different tasks over the feature map are similar, then it means that the two different tasks focus on the same region. Explorations by Lee et al. (2019) and Li et al. (2021) have shown that spatial correlations in the image often represent semantic correlations, and that effective auxiliary task branches can be designed based on such correlations to assist in the training of the main task. Therefore, we design $s^t_{a\to m}$ for capturing the semantic correlation between task branches.

For the problem that training set information cannot truly characterize the generalization change of model with few samples, the change of main task performance on validation set after using auxiliary tasks' information to optimize shared parameter is calculated as a supplementary information. The $z^t_{a\to m}$ in EAT is designed to capture generalization. $z^t_{a\to m}$ indicates from the results whether the introduction of the auxiliary task enabled the main task to achieve performance gains even on datasets not involved in training. At time step (one update of network parameters) t of training phase, EAT is computed as follows:

$$EAT = s^t_{a\to m} \cdot z^t_{a\to m} \tag{1}$$

$$\begin{aligned} s^t_{a\to m} &= \max\{0,\ \mathrm{sgn}(cs_{a\to m})\} \\ &= \max\left\{0,\ \mathrm{sgn}\left[\cos\left\langle \frac{\partial loss^t_m}{\partial f^t_s},\ \frac{\partial loss^t_a}{\partial f^t_s}\right\rangle\right]\right\} \end{aligned} \tag{2}$$

$$\begin{aligned} z^t_{a\to m} &= \max\{0,\ \mathrm{sgn}(TA_{a\to m})\} \\ &= \max\left\{0,\ \mathrm{sgn}\left[1 - \frac{loss_m\left(x^t_{val},\ \theta^{t+1}_{s|a},\ \theta^t_m\right)}{loss_m\left(x^t_{val},\ \theta^t_s,\ \theta^t_m\right)}\right]\right\} \end{aligned} \tag{3}$$

$\partial loss^t_m / \partial f^t_s$ is the gradient of shared feature map at time step$t$ caused by the backward of main task loss. $\partial loss^t_a / \partial f^t_s$ is the gradient of shared feature map at time step$t$ caused by the backward of an auxiliary task loss. $cs_{a\to m}$ is the cosine similarity of $\partial loss^t_m / \partial f^t_s$ and $\partial loss^t_a / \partial f^t_s$. sgn() is sign function. $TA_{a\to m}$ is the task affinity of the auxiliary task computed on the validation set based on the method proposed by Fifty et al. (2021).

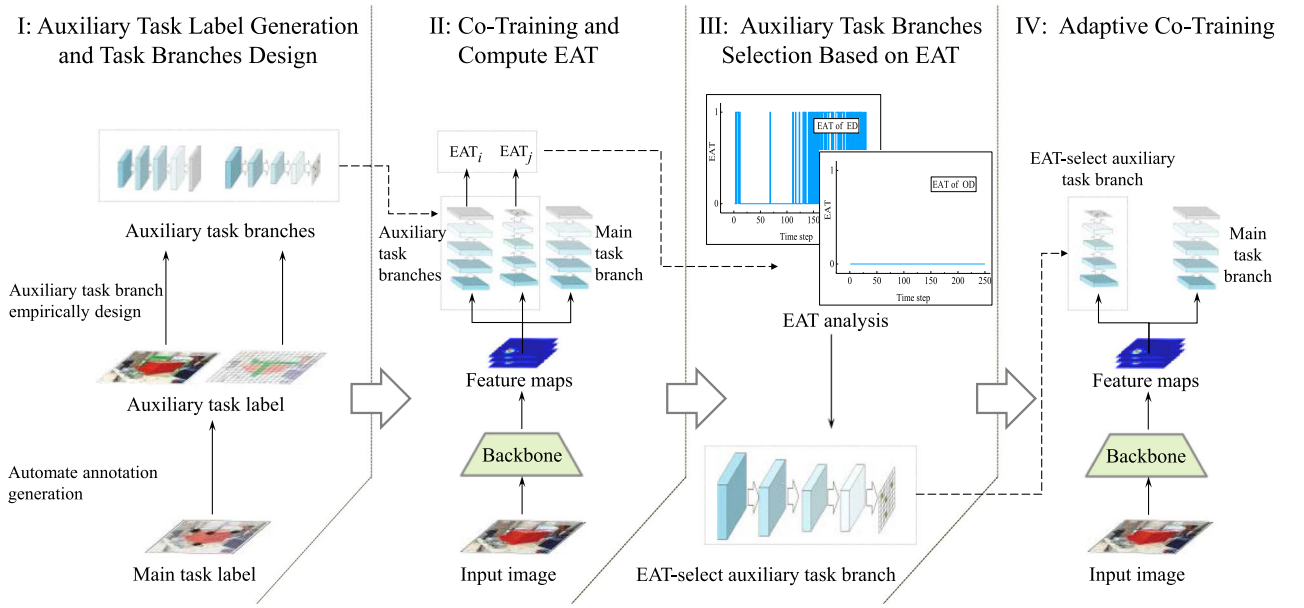

Fig. 2 Overview of proposed multi-task learning approach to vision tasks with few samples. (I): Generate auxiliary task labels and empirically design auxiliary task branches. (II): Co-train the empirically designed auxiliary task branches with the main task branches, and calculate the EAT of them. (III): Analyze the EAT of each auxiliary task branch and select potentially effective auxiliary task branches. (IV): Adaptively co-train the selected auxiliary task branches with the main task to improve the performance of the main task

If $s^t_{a\to m} = 1$ and $z^t_{a\to m} = 1$, then EAT = 1. The introduction of this auxiliary task branch is considered to contribute to the main task performance improvement at this time step. If $s^t_{a\to m} = 0$ or $z^t_{a\to m} = 0$, then EAT = 0. The introduction of this auxiliary task branch is not considered to contribute to the main task performance improvement at this time step.

EAT is designed with the following considerations: on the one hand, $s^t_{a\to m} = 1$ indicates that two task branches are focusing on similar regions of the shared feature map at this time step. On the other hand, according to the chain rule, the shared parameter update $\Delta\theta^t_m$ (caused by main task loss) and $\Delta\theta^t_a$ (caused by auxiliary task loss) are satisfied:

$$\Delta\theta^t_m = \frac{\partial loss^t_m}{\partial\theta^t} = \frac{\partial loss^t_m}{\partial f^t_s}\cdot\frac{\partial f^t_s}{\partial\theta^t} \tag{4}$$

$$\Delta\theta^t_a = \frac{\partial loss^t_a}{\partial\theta^t} = \frac{\partial loss^t_a}{\partial f^t_s}\cdot\frac{\partial f^t_s}{\partial\theta^t} \tag{5}$$

From the above equation, it can be known that the similarity between $\Delta\theta^t_m$ and $\Delta\theta^t_a$ depends on the similarity between $\partial loss^t_m/\partial f^t_s$ and $\partial loss^t_a/\partial f^t_s$. Therefore, when $s^t_{a\to m} = 1$, it indicates that the direction of parameter update caused by auxiliary task is similar to that caused by main task. It means the introduction of auxiliary task branch promotes the decrease of main task's training error. $z^t_{a\to m} = 1$ indicates that the introduction of auxiliary task branch can promote the decrease of main task's validation error. Therefore, EAT = 1 indicates that the introduction of auxiliary task at this time step can make both decrease of training error and validation errors. In other words, it makes model parameters to be updated in the direction of model generalization's improvement.

Automatic selection method of auxiliary task branches

The automatic selection method is proposed for the highly experience-dependent problem of manual selection. Alternative auxiliary task branches are co-training with the main task branch and their EAT is computed. During the training process, auxiliary task branches that consistently or periodically maintain an EAT value of 1 are considered as potentially effective auxiliary task branches. Conversely, if a task branch consistently maintains an EAT value of 0 during training, it signifies that the task branch does not improve the model generalization during the training process and is therefore discarded. Figure 3 shows proposed automatic selection method based on EAT.

It should be emphasized that the overall training process determines the final performance of the main task. While an auxiliary task branch may enhance the performance of the main task at certain time steps, it does not guarantee improvement in the final performance of the main task branch. This situation will be discussed in section "Commutator surface defect segmentation" through experimental analysis.

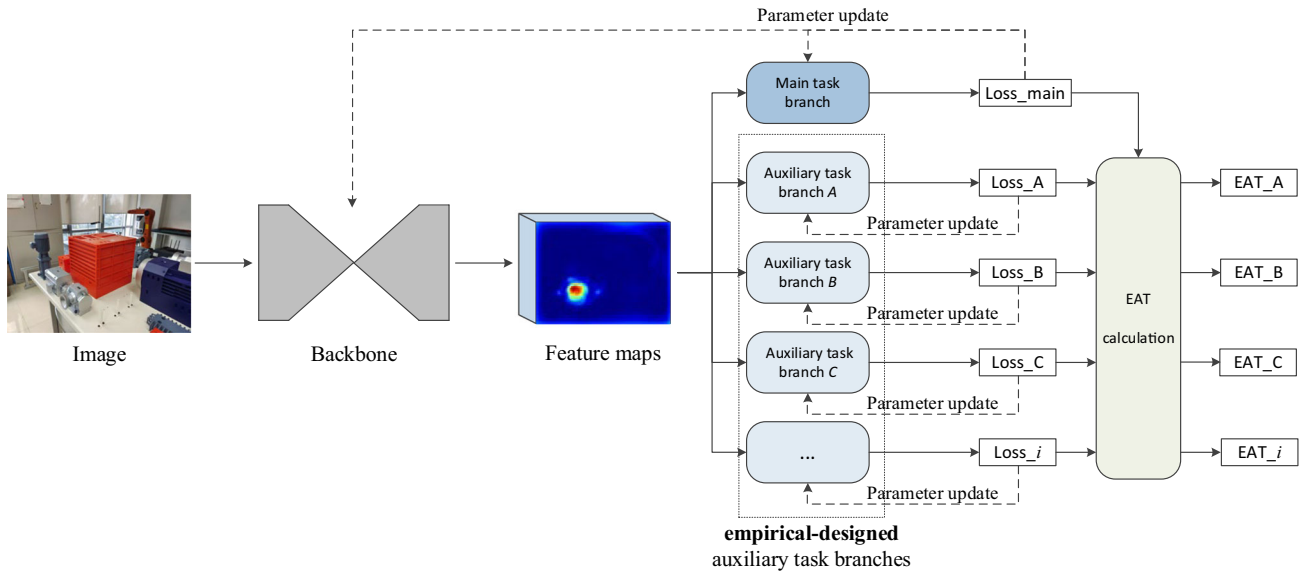

Fig. 3 The selection of Auxiliary task branches. The parameters of backbone and main task branch are updated by main task loss, and the parameters of auxiliary task branches are updated by their own losses

The proposed automatic selection method aims to select the auxiliary task branches that have demonstrated performance improvement for the main task during training stage. Auxiliary task branches that are excluded by this method are deemed ineffective, as they fail to contribute to the performance of the main task throughout the entire training process.

Adaptive co-training of multi-task branches

For the timing problem of auxiliary task information introduction, adaptive co-training method is proposed to train the selected auxiliary task branches with the main task branch. For each time step, shared parameters are updated by loss of main task branch and the loss of auxiliary task branches whose EAT = 1. Each task branches' parameters are updated by their own loss. Taking SGD optimizer as an example, for K auxiliary task branches, at time step t, proposed adaptive co-training method updates model parameters as following:

$$\Delta\theta^t = \frac{\partial loss_m^t}{\partial \theta^t} + \sum_i^K w_i \cdot EAT_i^t \cdot \frac{\partial loss_a^{i\,t}}{\partial \theta^t} \tag{6}$$

$$\Delta\sigma_m^t = \frac{\partial loss_m^t}{\partial \sigma_m^t} \tag{7}$$

$$\Delta\sigma_a^{i\,t} = \frac{\partial loss_a^{i\,t}}{\partial \sigma_a^{i\,t}} \tag{8}$$

$\Delta\theta^t$ is the update amount of the shared parameters. w_i is the weight coefficient of auxiliary task i. EAT_i^t is the EAT of auxiliary task i at time step t. σ_m^t is the branch's parameter of the main task. $\Delta\sigma_m^t$ is the branch's parameter update amount of the main task. $\sigma_a^{i\,t}$ is the branch's parameter of auxiliary task i. $\Delta\sigma_a^{i\,t}$ is the branch's parameter update amount of auxiliary task i. Figure 4 shows how to conduct adaptive co-training based on EAT.

The proposed adaptive co-training enables the auxiliary task branches to only provide information for shared parameters updating at the time step where they are beneficial for improving main task performance. At the time step of EAT = 0, the information provided by the auxiliary task branches do not participate in shared parameters updating. This effectively prevents any negative impact that the auxiliary task branches may have on the performance of the main task. Consequently, the proposed adaptive co-training method maximizes the potential of the auxiliary task branches to improve the performance of the main task.

After completing adaptive co-training, the inference process of the proposed multi-task approach is consistent with that of single task model. Only the backbone and the main task branch are used, without using other task branches or calculating EAT.

Experiments

Two vision tasks with few samples, product position keypoints detection and commutator surface defect segmentation, are used to validate proposed method. The samples of the former are from our annotation for manual assembly simulation scenarios and the samples of the latter are from the Kolektor Surface-Defect Dataset (Tabernik et al., 2020). We employed a workstation equipped with an i9-10900K CPU, 32 GB of RAM and RTX 3090 GPU for validations. The

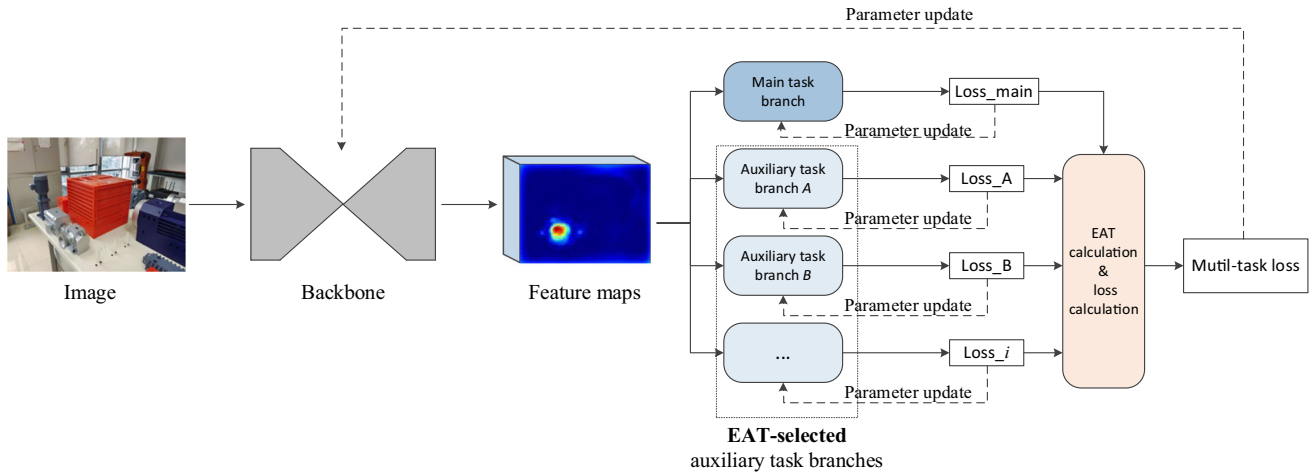

Fig. 4 Adaptive co-training based on EAT

validations were conducted on the Windows 10 operating system, with Python and PyTorch as the programming languages and the deep learning framework.

These two tasks have been selected for the validation of the proposed approach based on two main considerations. Firstly, high-quality manual annotations are required in these two tasks, and only few samples can be collected (only 52 surface defect annotations were collected in the Kolektor Surface-Defect Dataset). Secondly, these two tasks serve as representative examples of two different types of vision tasks encountered in industrial manufacturing. Commutator surface defect segmentation exemplifies the vision tasks characterized by a fixed view angle, a small view field and a simple background (like product appearance quality inspection in an assembly line). Product position key-points detection can represent the vision tasks characterized by an open view, a large view field and a complex background (like artificial satellite manual assembly).

Product position key-points detection

Product position key-points detection aims to estimate 2D pixel-coordinates of the key-points for subsequent product position estimation. It is an important pre-step for manufacturing processes such as robotic grasping (Xie et al., 2023). The dataset used in this paper is our self-annotated dataset for the simulation scenario of manual product assembly. The dataset contains images of a product in different viewpoints (as shown in Fig. 5a) and the 2D pixel coordinates of its four key-points. The average normalized pixel error (ANPE) is used to evaluate the performance of different methods on the product key-points detection task. The ANPE is a metric commonly used to evaluate the accuracy of pixel-wise predictions. It measures the average discrepancy between the predicted pixel values and the ground truth values, normalized by the size of the image.

Backbone and task branches design

We below describe the details of main task and four empirical-designed auxiliary task branches. The empirical-designed auxiliary tasks includes *key-points rough detection* (*KD*), *foreground classification* (*FC*), *edge detection* (*ED*) and *object detection* (*OD*). The form of label for each task branch is shown in Fig. 5.

Backbone Stacked hourglass model (Newell et al., 2016) is used as backbone.

Main task As the main task of key-points detection has few manual annotations, which mark the 2D coordinate of key-points in every image. The branch structure is shown in Fig. 6a. The number of output channels is the same as the number of key-points, and each output channel is the heatmap of corresponding key-points. MSE loss is used as the main task loss.

Key-points rough detection (KD) KD aims to localize the cell where the corresponding key-point is located from the 16×16 cells of input image. The structures of KD, FC and ED are shown in Fig. 6b. The number of output channels of KD is the same as the number of key-points, and the number of output channels of FC and ED is 1. Cross-entropy loss is computed as the task loss of KD. KD labels are generated from the main task labels of which cell the key-points are in.

Foreground classification (FC) FC aims to detect cells with key-points from the 16×16 cells of input image. The labels of FC (as shown in Fig. 5d) is also generated from the main task labels of which cell the key-points are in. Cross-entropy loss is computed as the task loss of FC.

Edge detection (ED) ED aims to detect the cells containing the edges from the 16×16 cells of input image. ED labels (as shown in Fig. 5e) is the cells through which the key-points are connected. Cross-entropy loss is computed as the task loss of ED.

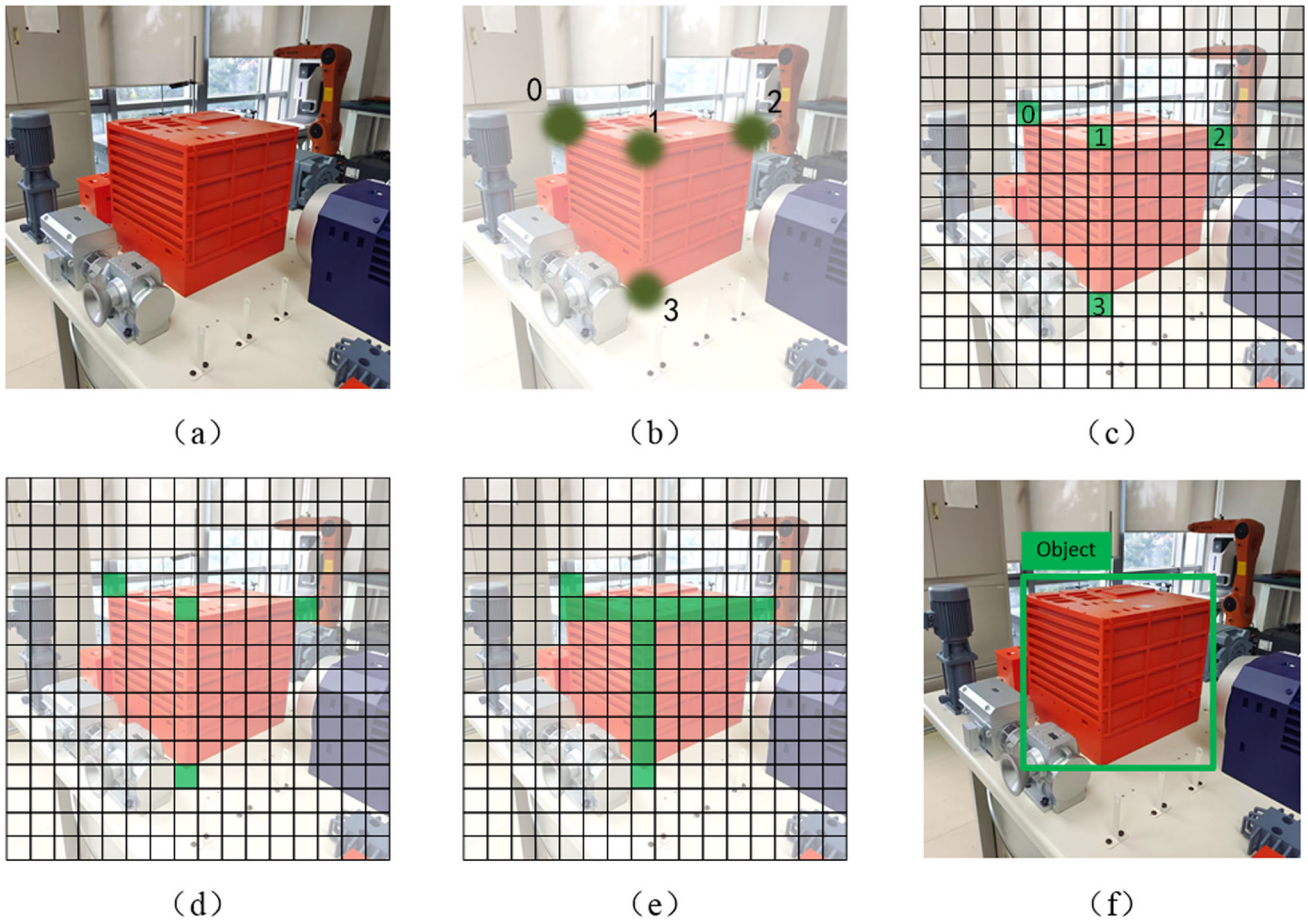

Fig. 5 Input image of key-points detection and labels corresponding to each task branch. **a** Input image. **b** Label of the main task. **c** Label of key-points rough detection. **d** Label of foreground classification. **e** Label of edge detection. **f** Label of object detection

Object detection (OD) OD aims to detect the product in the form of bounding box. The branching structure, labeling form and loss function of OD are consistent with those in YOLO (Redmon et al., 2016). The labels of OD are computed based on the coordinates of key-points.

Experiments and discussions

The training set used in the experiment contains 20 samples. The validation set contains 5 samples. The test set contains 30 samples. The EAT values of each auxiliary task branch in different training modes are shown in Fig. 7.

As shown in Fig. 7a, the stage of EAT = 1 exists in the co-training of FC, KD and ED. They are select as potentially effective auxiliary task branches according proposed automatic selection method. EAT of OD always is 0 in co-training, which is considered by proposed method to be unable to improve the main task performance. The selected auxiliary task branches (FC, KD and ED) are used to adaptive co-training with main task branch, and their EAT in the adaptive training process is shown in Fig. 7b.

Taking the average normalized pixel error (ANPE, the lower the ANPE, the better the method works) between key-points detection results and labels as statistical metric, the performance of main task obtained under different training modes and task combinations is shown in Table 1.

From Table 1, it shows that directly co-training of auxiliary task, which is considered ineffective by proposed method, with the main task will reduce the main task performance (like Main task + OD). Co-training the selected auxiliary tasks with the main task is also likely to reduce the main task performance (like Main task + FC). But in the adaptive co-training mode, the selected auxiliary task branches all improve main task performance, and the improvement effect is better than the co-training mode. It proves that the proposed method is able to remove invalid auxiliary task branches (like OD) and enhance the promotion effect of selected auxiliary task branches on main task performance. Therefore, the proposed method can improve the performance of the DL method for vision task with few samples.

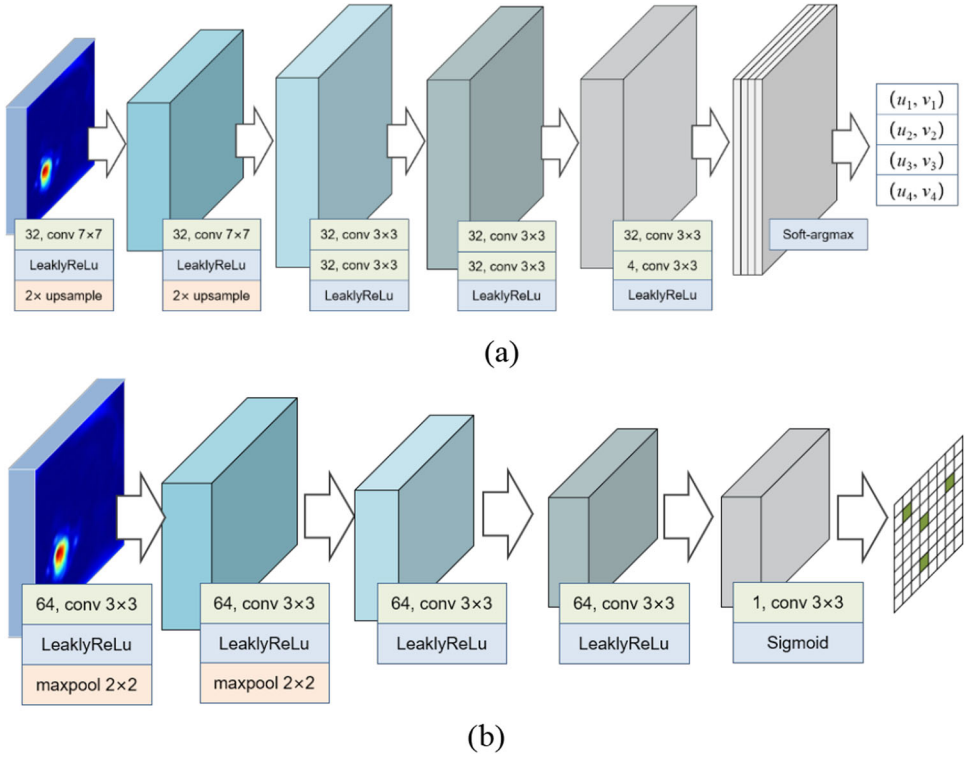

Fig. 6 Network structure corresponding to each task branch of key-points detection. **a** Structure of main task branch. **b** Structure of KD, FC and ED. The difference is in the number of their output channels

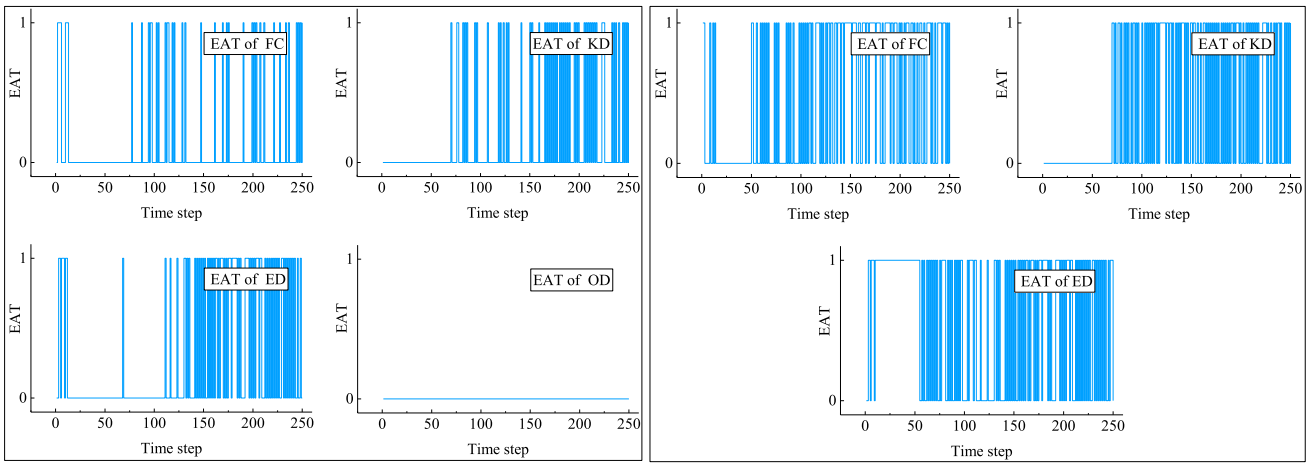

Fig. 7 EAT of auxiliary task branches during different training modes. **a** EAT during co-training. **b** EAT during adaptive co-training

Commutator surface defect segmentation

Commutator surface defect segmentation aims to detect surface cracks of electrical commutator at the pixel level for product quality inspection. The rarity of negative samples in the production process and the complexity of the labeling effort make it difficult to build large-scale defect datasets for this task. Only 52 annotations of surface defect images are contained in the public dataset Kolektor Surface-Defect Dataset. We would like to verify the effectiveness of proposed method with the help of this task and dataset. The Intersection over Union (IoU) is used to evaluate the performance of different methods on the commutator surface defect segmentation task. It quantifies the degree of overlap between the predicted segmentation mask and the ground truth mask of a surface defect. Mathematically, IoU is defined as the ratio of the area of intersection between the predicted and ground truth regions to the area of their union. A higher IoU value indicates a greater degree of overlap between the predicted and ground truth regions, implying better accuracy in segmentation. Conversely, a lower IoU suggests poorer alignment between the predicted and actual regions.

Table 1 Average normalized pixel error of different training mode and task combinations

Training modes	ANPE
Single main task	0.0613
Main task + FC	0.1217
Main task + FC (proposed method)	0.0396
Main task + ED	0.0443
Main task + ED (proposed method)	0.0220
Main task + KD	0.0232
Main task + KD (proposed method)	0.0207
Main task + OD	0.1659
Main task + FC + ED + KD	0.0195
Main task + FC + ED + KD (proposed method)	**0.0143**

Bold values indicate the best method
The input image is sized at 512 × 512 pixels

Backbone Stacked hourglass model is also used as backbone.

Main task The locations of the surface defects were labeled with a binary map (Fig. 8b). The branching structure is shown in Fig. 9a. Cross-entropy loss is computed as main task loss.

Defect rough location (DRL) DRL is designed to perform rough detection of regions where surface defects exist. The branching structure is the same as the main task. The labels are obtained from the main task labels by an expansion operation (Fig. 8c). The pixel value of the real defect is set to 1, while the pixel values of additional regions obtained by the expansion operation are set to 0.5. The MSE loss is computed as task loss.

Defect existence judgment (DEJ) The DEJ branch aims to perform a binary classification to discriminate whether the input image have cracks or not. The structure of the branch is shown in Fig. 9b. Cross-entropy loss is computed as task loss.

Backbone and task branches design

We below describe the details of main task and two empirical-designed auxiliary task branches: *defect rough location* (*DRL*) and *defect existence judgment* (*DEJ*). The former performs rough localization of defect in the input image, and the latter performs a binary classification to discriminate whether the input image have cracks or not. The input image, the label of main task and DRL are shown in Fig. 8.

Experiments and discussions

26 samples with surface defects and 26 samples without surface defects are contained in the training set. 5 samples with surface defects and 5 samples without surface defects are contained in the validation set. 21 samples with surface defects and 2 samples without surface defects are contained in the test set. different training modes of EAT is shown in Fig. 10.

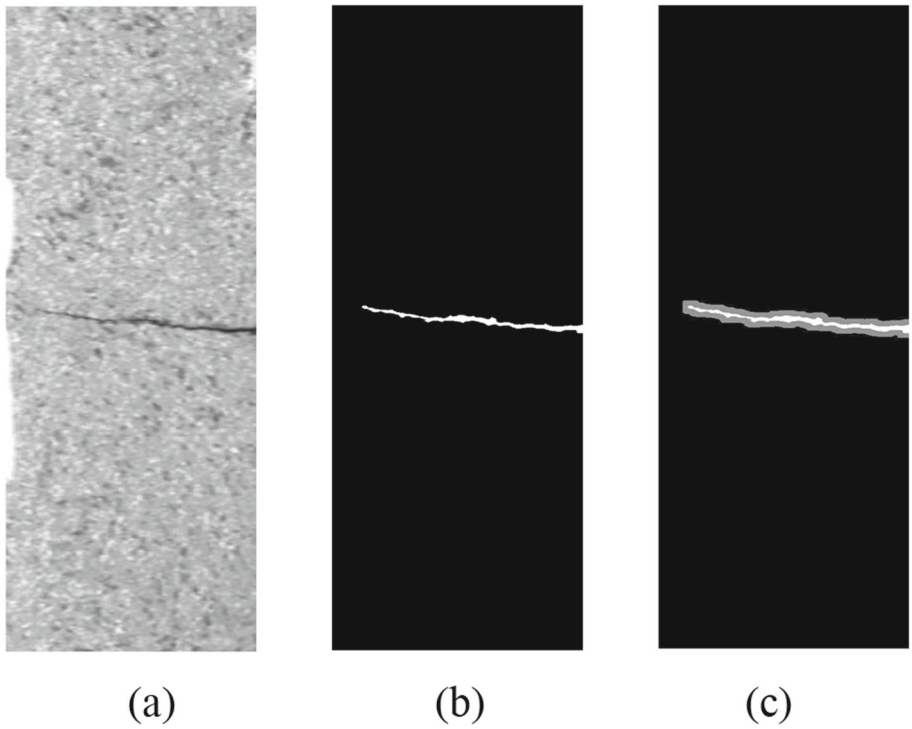

Fig. 8 Input image of commutator surface defect segmentation and labels of main task branch and defect rough location branch. **a** Input image. **b** Label of main task. **c** Label of defect rough location (DRL) branch

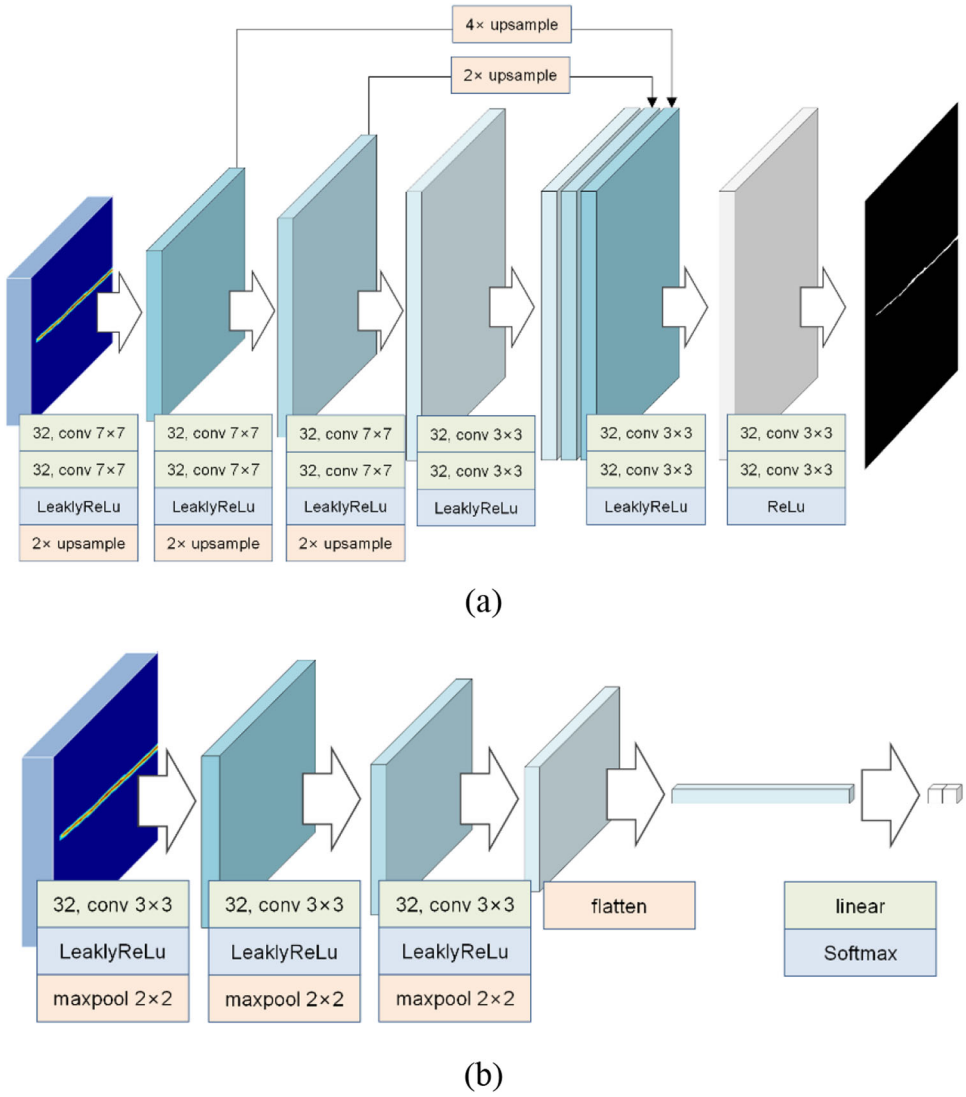

Fig. 9 Model structure corresponding to each task branch of surface defect segmentation. **a** Structure of main task branch and DRL branch. **b** Structure of DEJ branch

As shown in Fig. 10, both DRL and DEG in co-training have stages of EAT $\neq 0$. Both are selected as potentially effective auxiliary task branches. Although the changes of EAT are not same during different training modes, they indicated that DRL and DEG may be effective at the early stage of training process.

Taking the Intersection over Union (IoU, the higher IoU, the better the method works) of surface defect detection results and labels as a statistical metric, the defect segmentation effect of the model obtained for different combinations of training modes and tasks is shown in Table 2.

The experimental results indicate that proposed method has improved the performance of the main task. It is worth noting that the EAT value for the DRL branch is 1 during certain early stages of the training process, but it cannot improve the performance of the main task in co-training or adaptive co training. It's crucial to recognize that the overall training process ultimately determines the final performance of the main task. While an auxiliary task branch may contribute to the main task's performance at some stages, it doesn't guarantee improvement in the main task's final performance. This indicates that during other stages of the training process, the negative impact of the DRL branch on the main task outweighs its earlier promotion, resulting in a detrimental effect of the DRL branch on the final performance of the main task branch.

However, by employing the proposed method, the detrimental impact of DRL can be significantly mitigated (as seen in Main task + DRL (proposed method) and Main task + DRL + DEL (proposed method)), with the main task performance approaching the optimal level unaffected by DRL interference, such as 0.6383 for Single main task and 0.6339 for Main task + DRL (proposed method), 0.6927 for Main task + DEL (proposed method) and 0.6921 for Main task + DRL + DEL (proposed method). This underscores the advantages and necessity of the proposed adaptive co-training method: despite the inclusion of auxiliary task branches with negative impacts like DRL, it effectively mitigates these effects and enables strong main task performance.

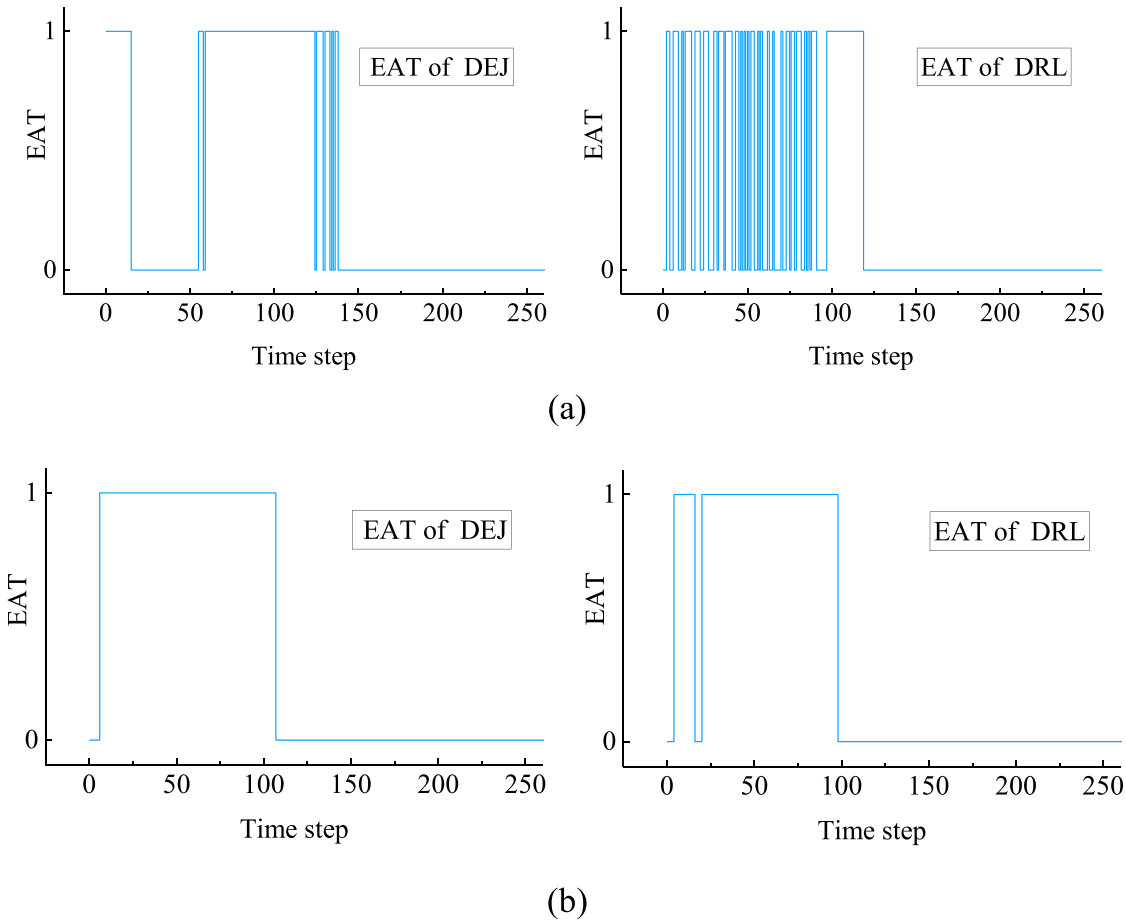

Fig. 10 EAT of DRL and DEJ during different training modes. **a** EAT during co-training. **b** EAT during adaptive co-training

Table 2 Defect segmentation performance on test set under different training modes

Training modes	IoU
Single main task	0.6383
Main task + DRL	0.0341
Main task + DRL (proposed method)	0.6339
Main task + DEJ	0.6591
Main task + DEL (proposed method)	**0.6927**
Main task + DRL + DEJ	0.0344
Main task + DRL + DEL(proposed method)	0.6921

Bold values indicate the best method

Comparative experiments

Comparison with other multi-tasking and meta-learning methods

This section presents a comparative analysis through experimental evaluation between the proposed methods and other prominent multi-tasking methods and meta-learning methods currently employed. We have selected GSM (Du et al., 2018), EITGM (Fifty et al., 2021), MTG-Net (Song et al., 2022), Auto-lambda (Liu et al., 2022) and STG-MTL (Sherif et al., 2023) for comparison.

GSM (Gradient Similarity Method) using cosine similarity between the gradients of the different tasks provides a signal to detect when an auxiliary loss is helpful to the main loss. EITGM (Efficiently Identifying Task Groupings Method) is an approach suggested by Fifty et al. (2021) to select which tasks should train together. The core of EITGM is a metric called inter-task affinity, which use the change in training error before and after co-training of two tasks to determine whether they should be trained together.

MTG-Net is a neural meta learner to capture these relationships between task combinations and performance gains. Auto-lambda is an automated weighting framework for exploring continuous, dynamic task relationships. STG-MTL is a Data Maps-based technique designed to group a large number of tasks. STG-MTL uses the soft grouping results as weights for multi-task loss balancing to further

improve task performance. These three techniques, originating from different technical paths such as meta learning, automated weighting framework, and Data Maps, respectively, attempt to address the task selection and co-training problems in multi-task learning and have achieved good results. We believe that these works represent the cutting-edge innovation currently.

The performance of different method is illustrated in Table 3.

From the experimental results, we can see that GSM wrongly selects two invalid auxiliary tasks, OD and DRL, in terms of task selection. EITGM, MTG-Net, Auto-lambda, and STG-MTL have missed to select FC, which is a valid auxiliary task, while the proposed method wrongly selects DRL, which is an invalid auxiliary task. Therefore, we believe that the proposed method reaches a level comparable to the three new methods MTG-Net, Auto-lambda, and STG-MTL in terms of task selection accuracy. And in terms of main task performance, the proposed method outperforms the three methods MTG-Net, Auto-lambda, and STG-MTL. Therefore, we believe that the proposed method has a comparative advantage.

Comparison with data augmentation and transfer learning methods

Data augmentation and transfer learning are also conventional scheme for few-shot learning. This section presents a comparative analysis through experimental evaluation between the proposed methods, data augmentation and transfer learning currently employed. Given that transfer learning necessitates source domains highly similar to the target domain and with sufficient supervision signals for effective knowledge transferring, we conducted comparative experiments using the Connector Surface defect segmentation task. The CrackForest dataset (Shi et al., 2016) was employed for the knowledge transfer process in transfer learning. Connector surface defects are similar to pavement cracks in CrackForest. The pavement crack images in the CrackForest dataset are shown in Fig. 11.

We specifically chose two innovative data augmentation methods, SFG (Zhang et al., 2022) and GLFA (Shi et al., 2023), along with two novel transfer learning approaches, iTFA (Choi & Kim, 2023) and G-FSDet (Zhang et al., 2023), for analysis and comparison with the proposed method. SFG (Sample-Centric Feature Generation) has devised a semi-supervised meta-generator to produce derivative features centered around each sample, thereby achieving data augmentation. GLFA (Global- and Local-aware Feature Augmentation) introduces a novel feature-level semantic augmentation, enhancing data augmentation from both channel and spatial dimensions. iTFA addresses catastrophic forgetting in transfer learning by employing incremental two-stage fine-tuning. G-FSDet enhances task performance by introducing metric-based discriminative loss into transfer learning process. The output structures of these methods have been modified based on the requirements of the tasks. The performance of different method is illustrated in Table 4.

The results of the experiments indicate that the performance of our proposed method performs slightly better than iTFA and significantly outperforms other methods. Notably, iTFA's success necessitates the supplementation of supervision signals from the CrackForest dataset. For diverse industrial vision tasks, it is often challenging to obtain a source domain with sufficiently supervised signals that are highly similar to the task's target domain. As a result, the application scope of transfer learning methods is severely constrained. In contrast, the proposed method does not require data from other domains and can be flexibly applied in various task scenarios. Considering both task performance and generalizability, we believe that our proposed method maintains a competitive advantage.

Ablation study and discussion

This section mainly conducts experimental discussions on six issues.

(a) Is the proposed method universal for different neural network models, or is it only effective for the Stacked hourglass model used in this experiment?
(b) $s^t_{a\to m}$ from training set and $z^t_{a\to m}$ from test set is used to form proposed metric EAT. Can better results be achieved by using training or testing set information alone for task selection and co-training?
(c) Proposed method uses a validation set to calculate $z^t_{a\to m}$ during the training process, then is the improvement in the main task performance only due to the use of validation set data during the training process?
(d) EAT is binarized. If we change EAT to a function that continuously increases with the increase of gradient similarity, can we further improve the effectiveness of proposed adaptive co-training method?
(e) How does the proposed multi-task learning approach affect the computational cost of the model?
(f) Can the proposed method be integrated with other methods to improve task performance even further?

For (a), the universality of proposed method is verified by replacing different backbones. For (b), $s^t_{a\to m}$ and $z^t_{a\to m}$ are used to task select and co-training, and the differences between them and EAT are compared. For (c), the validation set data are added to the training set for experimentation and compare the differences between this method and proposed method. For (d), three different EAT calculation methods

Table 3 Comparative experiments of multi-task learning and meta-learning methods

Method	Selected task of Key-points detection	Selected task of Surface defect segmentation	Main task performance	
			Training modes	ANPE/IoU
GSM	FC, ED, KD, OD	DRL, DEJ	Main task + FC	0.0815
			Main task + ED	0.0523
			Main task + KD	0.0377
			Main task + OD	0.0613
			Main task + DRL	0.4317
			Main task + DEJ	0.6927
EITGM	ED, KD	DEJ	Main task + ED	0.0443
			Main task + KD	0.0232
			Main task + DEJ	0.6591
MTG-Net	ED, KD	DEJ	Main task + ED	0.0441
			Main task + KD	0.0229
			Main task + DEJ	0.6609
Auto-lambda	ED, KD	DEJ	Main task + ED	0.0343
			Main task + KD	0.0221
			Main task + DEJ	0.6713
STG-MTL	ED, KD	DEJ	Main task + ED	0.0407
			Main task + KD	0.0215
			Main task + DEJ	0.6807
Proposed method	FC, ED, KD	DRL, DEJ	Main task + FC	0.0396
			Main task + ED	0.0220
			Main task + KD	0.0207
			Main task + DRL	0.0341
			Main task + DEJ	0.6927

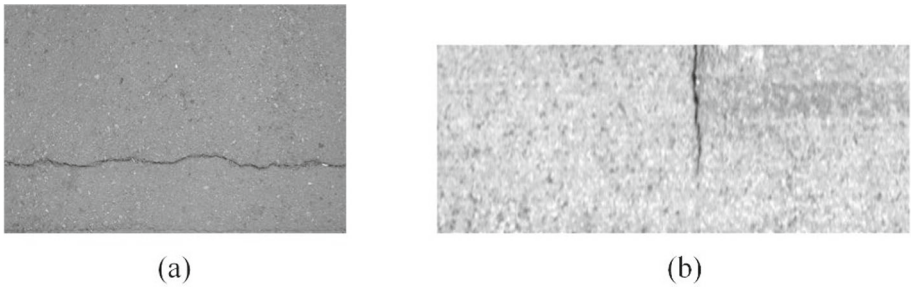

Fig. 11 Images of CrackForest and connector surface defect. **a** Pavement crack in CrackForest. **b** Connector surface defect

Table 4 Comparative experiments of data augmentation and transfer learning

Category of method	Method	Task performance (IoU)
Data augmentation	SFG	0.6471
	GLFA	0.6559
Transfer learning	iTFA	0.6912
	G-FSDet	0.6715
Multi-task learning	Proposed method (Main task + DEL)	0.6927

were designed and compared with proposed method to analyze the impact of EAT calculation methods. For €, the computational cost of the model under different training modes is calculated and compared. For (f), We have tried to integrate the proposed method with transfer learning methods.

Method universality analysis

Based on the key-points detection task, Resnet-18 (He et al., 2016) and VGG-11 (Simonyan & Zisserman, 2014) are used instead of the Stacked hourglass model as the backbone. Using Resnet-18 and VGG-11 as backbones for co-training and task select, the selected auxiliary task branches were consistent with the selected results using the Stacked hourglass model as backbones.

The results of key-points detection using different backbones are shown in Table 5. The results showed that the backbone structure would affect main task performance, but proposed method can still improve main task performance with different backbones. This can prove that proposed method is universal for different neural network models.

Furthermore, in order to validate the method of this paper on different datasets, we selected two additional datasets: railroad track surface defect dataset RSDDs (Gan et al., 2017) and the fabric defect detection database AFID (Silvestre-Blanes et al., 2019) for complementary experiments.

We use DRL and DEJ in Surface defect segmentation as auxiliary task branches for both RSDDs and AFID. Since there is no surface defect-free image in the RSDDs dataset, we cropped the defect-free part of the defective image and scaled it to the original size, which is used as the surface defect-free sample in DEJ. Stacked hourglass model is used as a backbone, and IoU is used as a task evaluation metric. The task performance on different datasets after experimentation using the method of this paper is shown in Table 6.

From the experimental results, it can be seen that the proposed method can improve the main task performance and enhance the facilitation of the auxiliary task to the main task. Another interesting aspect of the experimental results is that DRL is an invalid auxiliary task branch in Surface defect segmentation, which reduces the main task performance. However, in RSDDs, the DRL branch becomes an effective auxiliary task branch that enhances the main task. We believe this illustrates that the effectiveness of the auxiliary task is not only related to the model, but also affected by the task scenario and data distribution.

Analysis of metric differences

Based on the key-points detection task, $s^t_{a\to m}$ and $z^t_{a\to m}$ in co-training is calculated as shown in Fig. 12. With $s^t_{a\to m}$ and $z^t_{a\to m}$ replaces EAT for auxiliary task select using the same principle of EAT. It can be observed that the auxiliary task branches select results based on $s^t_{a\to m}$ are consistent with proposed method. But the select results based on $z^t_{a\to m}$ also suggest that OD may improve main task performance.

With $s^t_{a\to m}$ and $z^t_{a\to m}$ replaces EAT for adaptive co-training, and the main task performance is calculated as shown in Table 7. The experimental results show that the adaptive co-training based on $s^t_{a\to m}$ has achieved better performance than that it based on $z^t_{a\to m}$. But proposed method is better than both them. It proves that EAT have more advantages. The combination of training set information and validation set information is more conducive to multi task learning.

Impact analysis of using validation set data

Add the validation set data into the training set for co-training of auxiliary and main tasks, and the experimental results are shown in Table 8. The results show that the main task performance under the proposed adaptive co-training mode is better than that under the direct co-training mode using validation set data additionally. This proves that the reason of proposed method's improving main task performance is not only because the using of validation set data during the training process.

In addition, the experimental results also showed that proposed method can effectively reduce the negative impact of DRL on main task performance. Adding validation set data into the training set cannot solve this problem, and inappropriate auxiliary task branches can still have seriously negative impacts. This also demonstrates the advantages of proposed adaptive co-training method.

Table 5 Average normalized pixel error of using different backbone

Training modes	Resnet-18 backbone	VGG-11 backbone
Single main task	0.1463	0.1302
Main task + FC	0.0557	0.1137
Main task + FC (proposed method)	0.0306	0.0547
Main task + ED	0.0763	0.0858
Main task + ED (proposed method)	0.0192	0.0672
Main task + KD	0.1051	0.0370
Main task + KD (proposed method)	0.0561	0.0215

Table 6 Performance of different dataset

Dataset	Task branches selected by EAT	Training modes	Single main task	Co-training	Proposed method
Key-points detection	FC, ED, KD	Main task + FC	0.0613	0.1217	0.0396
		Main task + ED		0.0443	0.0220
		Main task + KD		0.0232	0.0207
Surface defect segmentation	DEJ, DRL	Main task + DEJ	0.6383	0.6591	0.6927
		Main task + DRL		0.0341	0.6339
RSDDs	DEJ, DRL	Main task + DEJ	0.5741	0.5814	0.6107
		Main task + DRL		0.6077	0.6257
AFID	DEJ, DRL	Main task + DEJ	0.4633	0.4911	0.5163
		Main task + DRL		0.0761	0.4582

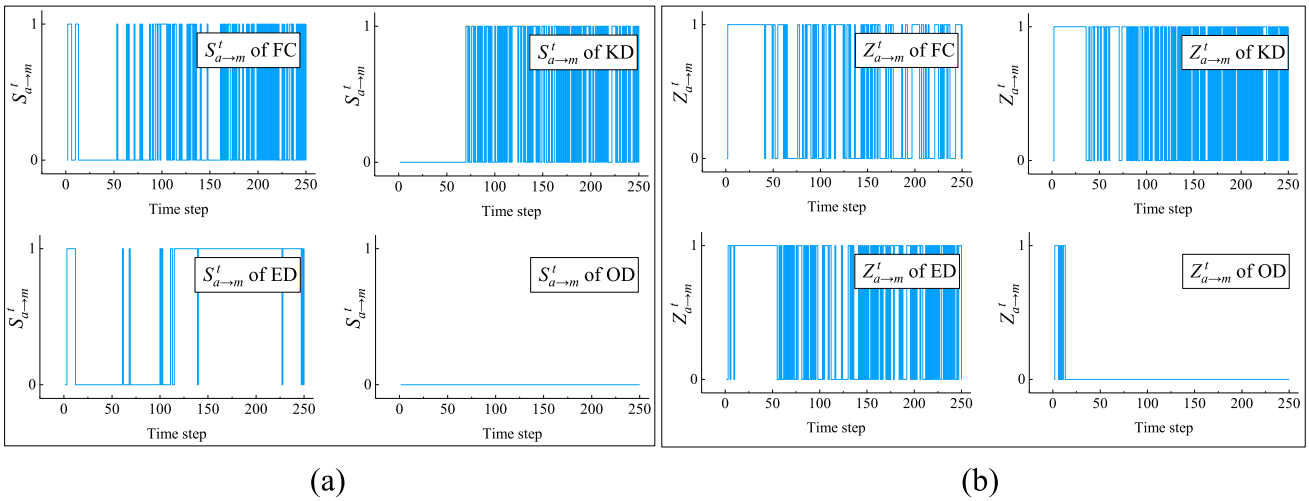

Fig. 12 $s^t_{a\to m}$ and $z^t_{a\to m}$ of auxiliary task branches in co-training. **a** $s^t_{a\to m}$ of auxiliary task branches in co-training; **b** $z^t_{a\to m}$ of auxiliary task branches in co-training

Influence analysis of EAT calculation method

We conduct the experiment by using sgn function and three representative smoothing functions commonly used in deep learning, *linear function*, *S function*, and *Inverse S function* (as shown in Fig. 13), for the calculation of $s^t_{a\to m}$ and $z^t_{a\to m}$ in EAT.

Using different functions to compute $s^t_{a\to m}$ and $z^t_{a\to m}$ and adaptive co-training with them, the best task performance of Key-points detection and Surface defect segmentation is shown in Table 9.

The results show that different EAT calculation methods can lead to differences in main task performance. But this difference is too small and does not demonstrate a comparative advantage of any calculation method. Therefore, we believe

Table 7 Average normalized pixel error of using different guiding metrics

Training modes	ANPE
Single main task	0.0613
Main task + FC (proposed method)	0.0396
Main task + FC (Guided by $s^t_{a\to m}$)	0.0445
Main task + FC (Guided by $z^t_{a\to m}$)	0.0649
Main task + ED (proposed method)	0.0220
Main task + ED (Guided by $s^t_{a\to m}$)	0.0362
Main task + ED (Guided by $z^t_{a\to m}$)	0.0520
Main task + KD (proposed method)	0.0207
Main task + KD (Guided by $s^t_{a\to m}$)	0.0233
Main task + KD (Guided by $z^t_{a\to m}$)	0.0767
Main task + OD (Guided by $z^t_{a\to m}$)	0.1048

Table 8 Main task performance of using different training set

Main task	Training modes	ANPE/IoU
Key-points detection	Main task + FC (proposed method)	0.0396
	Main task + FC (using validation set)	0.0801
	Main task + ED (proposed method)	0.0220
	Main task + ED (using validation set)	0.0403
	Main task + KD (proposed method)	0.0207
	Main task + KD (using validation set)	0.0281
Surface defect segmentation	Main task + DRL (proposed method)	0.6339
	Main task + DRL (using validation set)	0.0344
	Main task + DEL (proposed method)	0.6927
	Main task + DEL (using validation set)	0.6696

that using the simpler EAT calculation method proposed in this paper is acceptable.

Analysis of computational costs

To compare the differences in computational costs between proposed multi-task learning approach and the single task model, the floating-point operations (FLOPs) of them at different stages are calculated as shown in Table 10. Compared to the single task model, proposed multi-task learning approach requires additional computational overhead during the training phase. But in the inference stage, the computational cost of proposed multi-task learning approach is consistent with that of single task model. Therefore, proposed multi-task learning approach only increases the hardware cost of model training, without increasing the hardware cost of model deployment in manufacturing process.

In order to reduce the computational overhead, we try to use a larger time step $\boldsymbol{n}$ for the EAT calculation. However, too large a value of $\boldsymbol{n}$ may also reduce the final task effectiveness. In order to verify the effect of $\boldsymbol{n}$ value on the computational overhead and task effectiveness. We supplemented experiments and the experimental results are shown in Tables 11 and 12.

From the experimental results, it can be seen that increasing the $\boldsymbol{n}$ value can effectively reduce the computational overhead, but it also brings about a decrease in task performance. But the degree of performance degradation is not consistent across tasks when the $\boldsymbol{n}$ value is increased (like Main task + DEJ).

Integration with transfer learning

For the Surface defect segmentation task, we use the pavement crack dataset CrackForest (Shi et al., 2016) for transfer learning. First, we pre-train the backbone based on CrackForest, and use the supervised signals in CrackForest to optimize the parameters of the backbone initially. And then the pre-trained backbone is used as the backbone for the Surface defect segmentation task to execute the multi-task learning method in this paper. The final task performance is shown in Table 13.

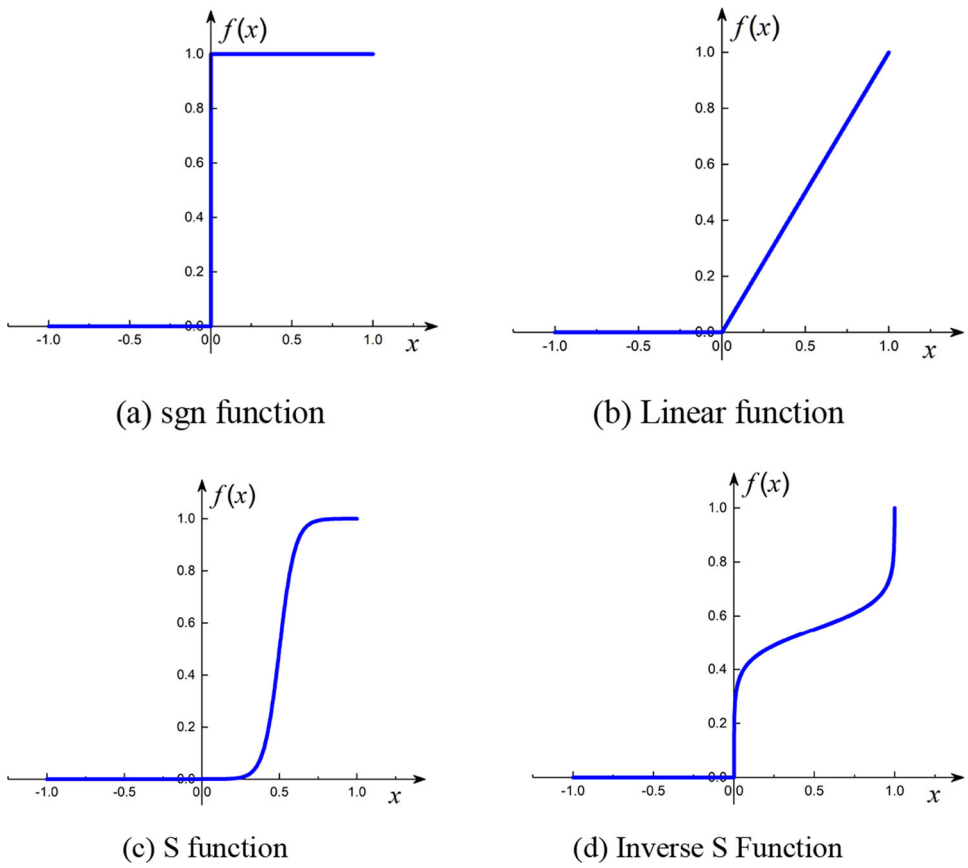

Fig. 13 Different calculation methods of EAT and their function image

Table 9 Best main task performance under different EAT calculation methods

Calculation method of $s^t_{a\to m}$	Calculation method of $z^t_{a\to m}$	Key-points detection (ANPE)	Surface defect segmentation (IoU)
sgn	sgn	0.0143	0.6927
sgn	Linear function	0.0148	0.6871
sgn	S function	0.0147	0.6947
sgn	Inverse S Function	0.0145	0.6940
Linear function	sgn	0.0141	0.6884
Linear function	Linear function	0.0151	0.6867
Linear function	S function	0.0144	0.6921
Linear function	Inverse S Function	0.0140	0.6931
S function	sgn	0.0145	0.6894
S function	Linear function	0.0149	0.6915
S function	S function	0.0143	0.6931
S function	Inverse S Function	0.0146	0.6899
Inverse S Function	sgn	0.0141	0.6907
Inverse S Function	Linear function	0.0147	0.6917
Inverse S Function	S function	0.0141	0.6931
Inverse S Function	Inverse S Function	0.0144	0.6913

Table 10 Computing overhead of proposed method and single-task model

Main task	Stage	FLOPs	Main task	Stage	FLOPs
Key-points detection	Co-training of proposed method	13.97 G	Surface defect segmentation	Co-training of proposed method	24.39 G
	Adaptive co-training of proposed method	12.86 G		Adaptive co-training of proposed method	24.39 G
	Inference of proposed method	6.82 G		Inference of proposed method	16.12 G
	Training of single-task model	6.82 G		Training of single-task model	16.12 G
	Inference of single-task model	6.82 G		Inference of single-task model	16.12 G

Table 11 Computational overhead (FLOPs) with different values of n

Main task	Stage	$n = 1$ (proposed method)	$n = 3$	$n = 5$	$n = 10$
Key-points detection	Co-training of proposed method	13.97 G	12.24 G	11.65 G	10.89 G
	Adaptive co-training of proposed method	12.86 G	11.83 G	10.91 G	10.12 G
Surface defect segmentation	Co-training of proposed method	24.39 G	21.19 G	20.75 G	20.44 G
	Adaptive co-training of proposed method	24.39 G	21.19 G	20.75 G	20.44 G

Table 12 Main task performance with different values of n

Main task	Training modes	$n = 1$ (proposed method)	$n = 3$	$n = 5$	$n = 10$
Key-points detection	Main task + FC	0.0396	0.0712	0.0933	0.1129
	Main task + ED	0.0220	0.0279	0.0331	0.0389
	Main task + KD	0.0207	0.0221	0.0228	0.0231
Surface defect segmentation	Main task + DEJ	0.6927	0.6915	0.6719	0.6657

Table 13 Performance of the proposed method integrating with transfer learning

Method	Training modes	IoU
Proposed method	Main task + DRL	0.0341
	Main task + DRL (proposed method)	0.6339
	Main task + DEJ	0.6591
	Main task + DEL (proposed method)	0.6927
Proposed method + transfer learning	Main task + DRL	0.0341
	Main task + DRL (proposed method)	0.6483
	Main task + DEJ	0.6714
	Main task + DEL (proposed method)	0.7017

As can be seen from the experimental results, the integration of transfer learning with the proposed method brings about an improvement in performance. Fusion of transfer learning, the task performance of Main task + DRL is still very poor. In contrast, the task performance of Main task + DRL (proposed method) improves after the integration of transfer learning. This proves that the negative impact of unsuitable auxiliary tasks on the main task is serious, and also proves the value of the adaptive co-training method proposed this paper from the opposite side.

Conclusion

In this paper, we present a novel multi-task learning approach to tackle the challenge of poor performance in DL methods

when dealing with vision tasks with few samples. Our proposed approach treats vision tasks as the main task in the multi-task learning paradigm. By leveraging a metric called Effective Auxiliary Task (EAT), we automatically select auxiliary task branches and conduct adaptive co-training. Experiments on vision tasks with few samples have shown that even without additional manual annotation, proposed approach can improve the performance of various neural networks for vision tasks with few samples.

In the future, our research will expand in various directions. Firstly, we will focus on mitigating the additional computational overhead arising from task selection and adaptive co-training, which holds paramount importance. Model compression algorithms and adaptive tuning of $\boldsymbol{n}$ will be the focus of our attempts. Secondly, we will investigate automated techniques for designing auxiliary task branches and developing adaptive optimization methods for the weighting of auxiliary task losses. Meta-learning or reinforcement learning approaches are promising directions. Additionally, we will further explore the integration with other few-shot learning techniques, including domain migration techniques and data augmentation techniques. Those integration may lead to better task performance.

Data availability The experimental dataset utilized in this work is sourced from five datasets—the publicly accessible Kolektor Surface-Defect Dataset, RSDDs, AFID, CrackForest and our privately annotated dataset. The former four datasets are available on their open source websites, while the latter will be made available on request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical approval Not applicable.

Consent to participant Not applicable.

Consent for publication Not applicable.

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